Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

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Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold

Warm-up: average variable or average total cost?

A workshop sews hoodies: total cost in euros a week

hoodies	0	25	50	75
total cost	400	700	900	1,300

At 0 hoodies it pays only the rent.

1. Average variable cost at 50 hoodies?
2. Average total cost at 50 hoodies?
3. The rent rises by 100 a week.
   Average variable cost at 50 now?

In pairs, two minutes: three answers; then hands.

Variable cost = total cost − fixed cost (class 11, slides 9–11).

Warm-up: which events shift demand to the right?

The demand for doughnuts in a town

1. Muffins, a substitute, become more expensive
2. Doughnuts get cheaper, and more of them are sold
3. Coffee, a complement, gets cheaper
4. The doughnut makers pay less for flour

In pairs, two minutes: four answers; then hands.

Its own price: a move along demand; related goods' prices, incomes, tastes: a shift (class 5, slides 13–14).

Madrid–Barcelona by fast train: one company, then three

Revealed in class

Before class: Ouigo (May 2021) and Iryo (November 2022) started running rival trains on this line. Guess what had happened by 2025 to the average fare and to the number of passengers — and why.

Source given in class.

From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	10	9	8	7	6
rides a day	100	200	300	400	500	600	700	800

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).

A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

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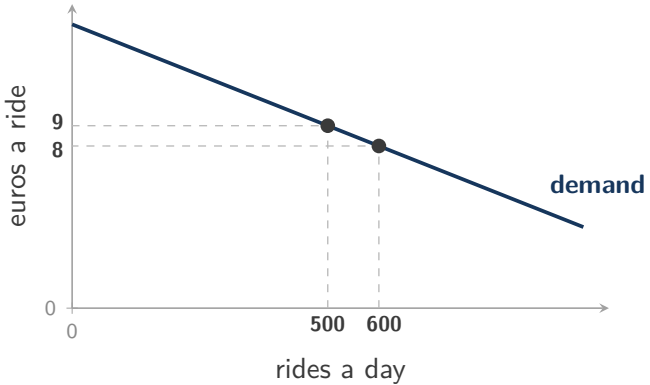
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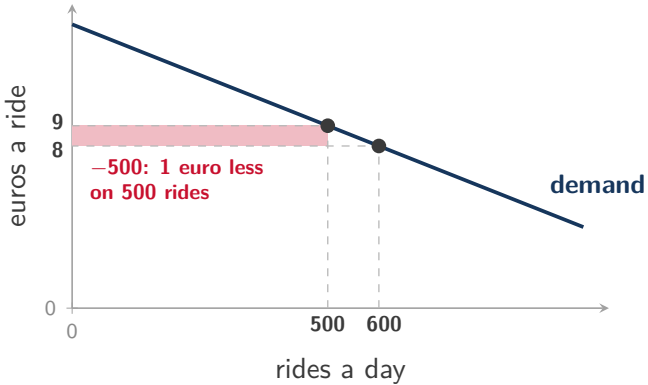
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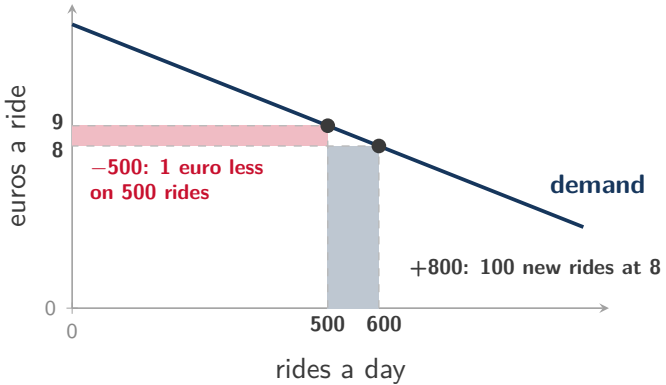
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



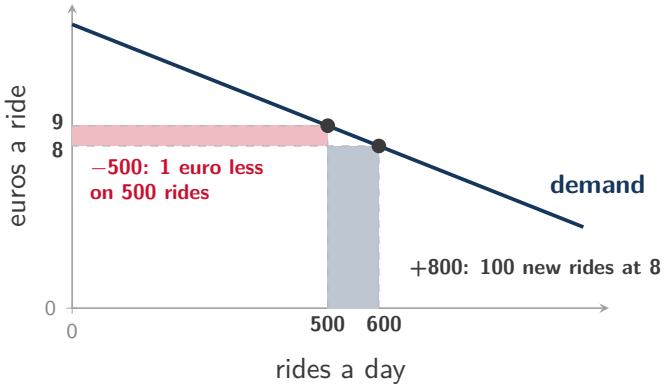
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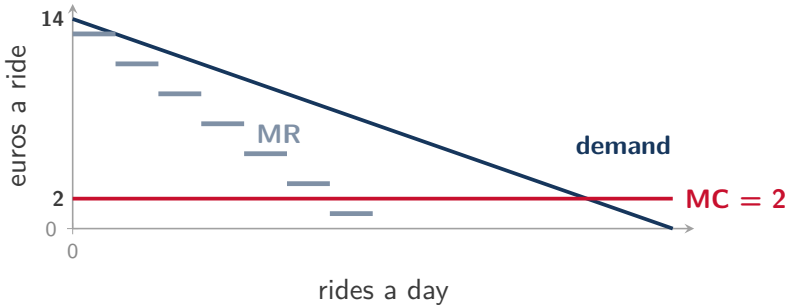
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: $800 - 500 = 300$ for 100 rides — 3 a ride, below the price of 8.

Which rides does the cable car sell?

Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



Before class: which is the last hundred whose marginal revenue beats the 2-euro cost? What price goes with it?

Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

What does the cable car's choice cost?

Revealed in class

Before class: at the cable car's price, some riders who value a ride at more than its 2-euro cost do not ride. How many rides would be sold at a price of 2? Who loses what? (The demand line: price $= 14 -$ rides $\div 100$.)

In pairs, four minutes: the only fibre network in town

One firm owns the town's only fibre network; each extra household costs it 10 euros a month.

highest price for all (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

1. Marginal revenue of each extra hundred, per household?
2. Its price, and how many households?
3. Its profit before the fixed cost?
4. How many households should get fibre, town and firm counted together?

In pairs, four minutes: four answers.

Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

The cable car again: which curve moves? (optional)

The cable car (slides 6–8): price $= 14 -$ rides $\div 100$; marginal revenue of each extra hundred: 13, 11, 9, 7, 5, 3, 1 a ride.

1. It raises its price from 10 to 12. Which curve moves?

2. A new road to the summit opens. What happens to its demand?

3. Each ride now costs 4, not 2. Which curve moves?

In pairs, two minutes: three answers.

A price change moves along the demand curve (class 5, slide 10).

Break

Ten-minute break

A different price for each rider?

Revealed in class

Price discrimination: selling the same good at different prices to different buyers.
Before class: if every rider paid exactly what the ride is worth to them, how many rides would the cable car sell, and who would gain the surplus?

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing

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In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

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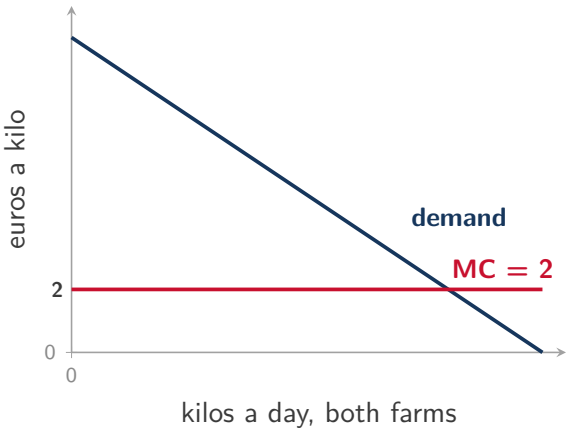
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- **Regulation:** cap the price a natural monopoly may charge
- **Public ownership:** the state runs it
- **Doing nothing:** when every cure costs more

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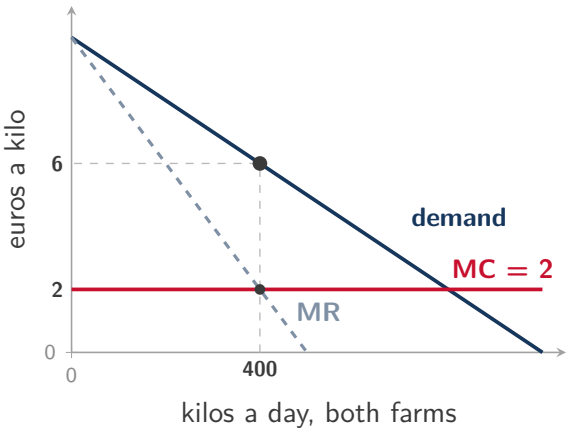
Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

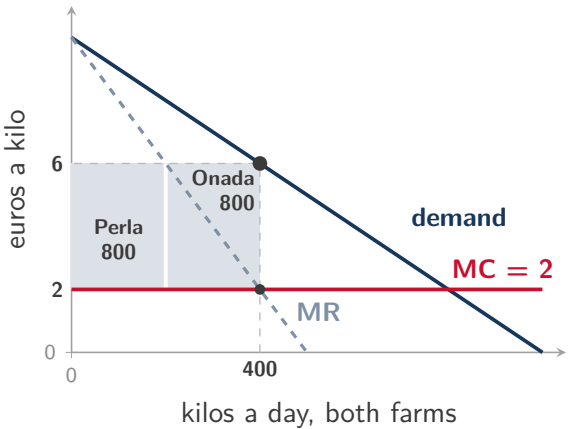
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Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Keep the deal or sell more? Each farm's profit a day

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos $\rightarrow$ 6.00 · 450 $\rightarrow$ 5.50 · 500 $\rightarrow$ 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

Where do the two farms end up?

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

In pairs, three minutes: open on Sundays?

A town's only two supermarkets: open on Sundays or not? Profit, thousand euros a month:

		Rodó	
		closed	open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

1. Fresc's best if Rodó stays closed?
2. And if Rodó opens?
3. Rodó's best choice?
4. The Nash equilibrium? Who gains?

In pairs, three minutes: four answers; then hands.

Revealed in class

Before class: as one monopoly the two farms sell 400 kilos at 6 euros; each kilo costs 2. Where does the price go when each farm does its best for itself, and when many farms share the bay? Why?

Meeting every day: can the deal hold?

Revealed in class

Before class: Perla breaks the deal on day 1, and from day 2 both sell 250 kilos every day. How much does Perla gain on day 1, and lose on each day after? Tit-for-tat: start with the deal, then do what the other did yesterday.

In pairs, three minutes: a stall at the town fair?

At the town fair, a toy chain, Mega, and a small toy shop, Petita, each choose a stall or not. Profit that day, thousand euros:

		Petita	
		stall	no stall
Mega	stall	Mega 9, Petita 2	Mega 10, Petita 3
	no stall	Mega 7, Petita 5	Mega 8, Petita 4

1. Does Mega have a dominant strategy?
2. Petita's best reply to each of Mega's?
3. The Nash equilibrium?

In pairs, three minutes: three answers.

Cartels and train fares: positive or normative? (optional)

1. *"Cartels raise the prices buyers pay."*
2. *"Firms in a cartel should pay bigger fines."*
3. A and B both want cheap train tickets. A: *"more companies on every line."* B: *"one public company with capped fares."* Values or a prediction?

In pairs, two minutes: three answers.

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

A season pass: what does one more trip cost? (optional)

Berta bought a cable-car season pass in October: 150 euros, no refunds. A trip takes an hour she could spend on a lifeguard shift paid 15 euros.

1. Her opportunity cost of one more trip today?
2. Should she ride more "to get her money's worth"?
3. Next autumn: is a new pass's 150 part of her choice?

In pairs, two minutes: three answers.

Opportunity cost: what you give up; sunk: paid and gone whatever you choose (class 2, slide 23).

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: bring your worked solutions on Thursday 12 November (not handed in)
- Practise in Sextant: the units the course page lists



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